Descriptive Statistics

Sunday, October 9, 2022 7:37 PM

* MEASURES OF CENTRAL TENDENCY :-

1) MEAN : AVERAGE

$$\mu = \frac{\text{Sum of all terms}}{\text{No. of terms}}$$

$$\boxed{\mu = \frac{1}{n} \sum_{i}^{n} x_i}$$

2) MEDIAN : Middle value

(i) n is odd : $\frac{n+1}{2}$ th

$2, 3, 4, \underline{5}, 6, 7, 8$ — 4th term

(ii) n is even : $\dfrac{\frac{n}{2}\text{th} + \frac{n}{2}+1\text{ th}}{2}$

$2, 3, 4, \underline{5, 6}, 7, 8, 9$

$\underline{5.5}$

3) MODE : The most frequent term

* EFFECT OF OUTLIER ON MEAN & MEDIAN :-

* MEAN :

$10, 20, 30, 40, 50 \qquad \mu = 30$

$10, 20, 30, 40, 500 \qquad \mu = 120$

> Mean shifts towards the outlier

* MEDIAN:

10, 20, 30, 40, 50 Median = 30

10, 20, 30, 40, 500 Median = 30

> Median is not affected by outliers

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* MEASURES OF SPREAD/ DISPERSION:-

(i) RANGE : UB - LB

(ii) VARIANCE :

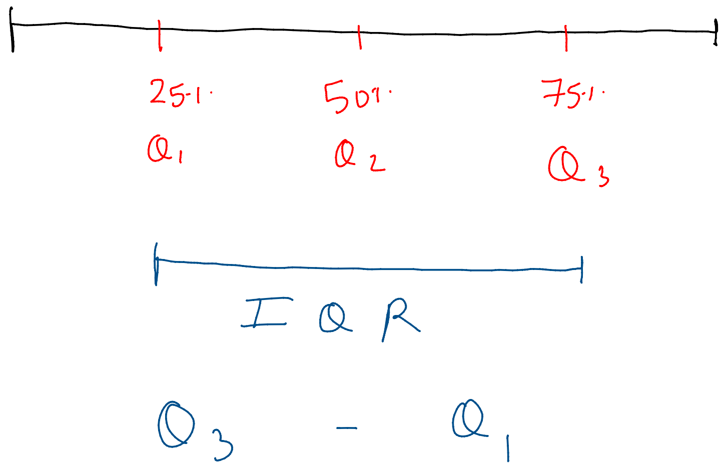
$$\boxed{\sigma^2 = \frac{1}{n} \sum_{i}^{n} (x_i - \mu)^2}$$

(iii) STANDARD DEVIATION:-

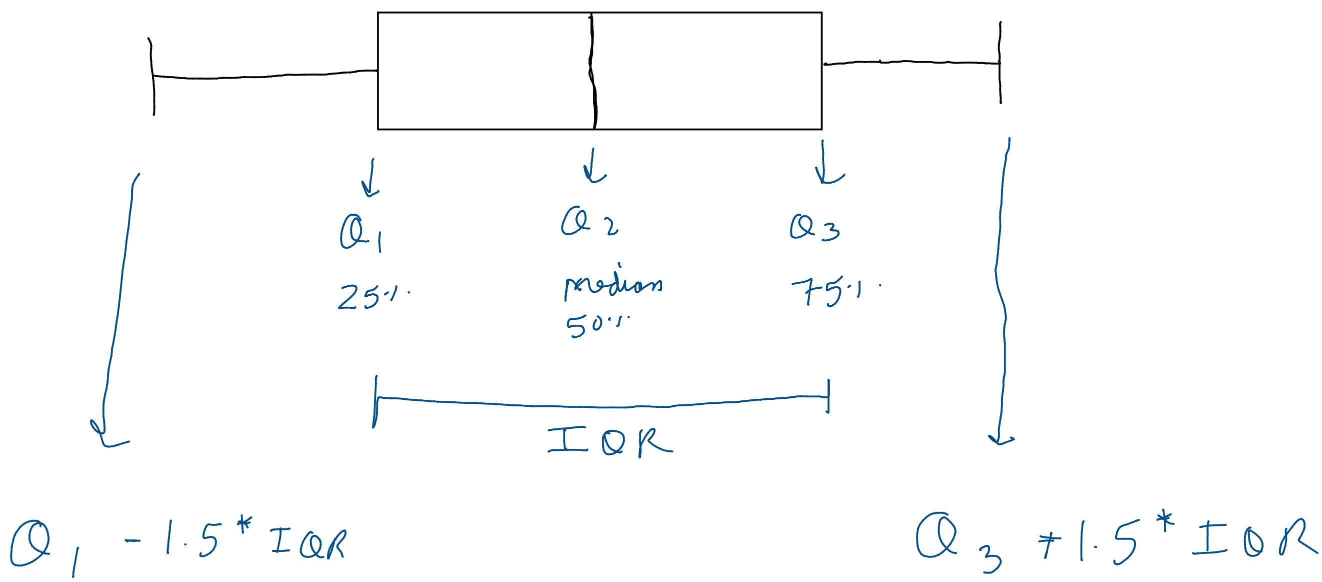
$$\boxed{\sigma = \sqrt{\frac{1}{n} \Sigma (x_i - \mu)^2}}$$

(iv) QUARTILES:-

> Quartiles divide our data into 4 quarters.



* READING THE BOX PLOT :-



* OUTLIER DETECTION :-

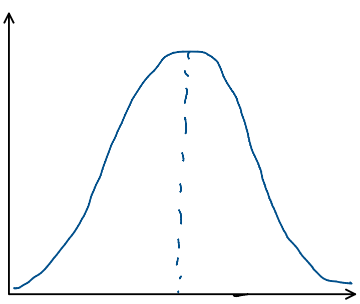
(i) Using Box Plots

(ii) Using z scores

$$z = \frac{x - \mu}{\sigma}$$

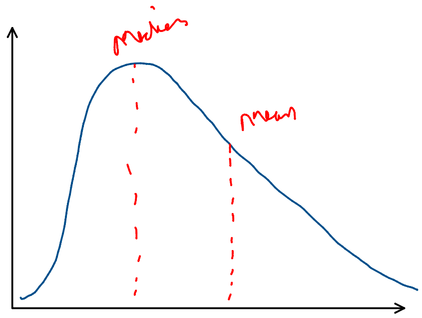
> Outliers have higher z scores

* SKEWNESS :-



Symmetric

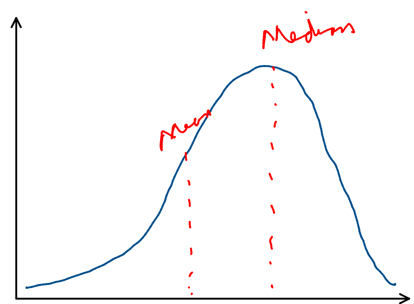
mean ≈ median



Right Skewed

Mean > median

Positive Skewed



Left Skewed

mean < median

Negative Skewed

* COVARIANCE
* CORRELATION

} Later

* COVARIANCE :-

> measure of linear association b/w two variables

$$\boxed{Cov(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n}}$$

* CORRELATION :-

$$\boxed{COR(x,y) = \frac{Cov(x,y)}{\sigma_x \sigma_y}}$$

$Cor(x,y) \simeq -1$: Very high negative correlation

$Cor(x,y) \simeq +1$: Very high positive correlation

Range : -1 to 1